

Document 3: Dynamic Parameter Chunking & Evaluation Specification

3.1 Parameter Description Layout

Each pedal instance registers its modifiable parameters via structured descriptors containing a unique identification token, an absolute minimum DSP scale value, and an absolute maximum DSP scale value.

3.2 Dynamic Runtime Chunk Slicing

The compiler determines parameter chunking on-the-fly based on the number of registered parameters. The total pool of cells within a pedal's assigned row boundaries is divided evenly among its parameters. Chunks are allocated sequentially using a deterministic scan order (Left → Right, Top → Bottom) confined within the pedal's row boundaries.

3.3 Chunk Evaluation Algorithm

Each chunk calculates a single baseline output value using color-weighted positional accumulation calculated row-by-row across its cells.

Color Contribution Weights: White (+2), Red (+1), Green (0), Blue (-1), Black (-2).

Positional Accumulation Logic: The algorithm steps through the cells of the chunk sequentially. If the cell index within the chunk is even, the signed color value is added to the accumulator. If the cell index is odd, the signed color value is subtracted from the accumulator.

Normalization and Range Mapping: Let A_{max} represent the maximum possible mathematical value achievable if every positive parity cell matches +2 and every negative parity cell matches -2. Let A_{min} represent the inverse absolute minimum. The value is normalized and scaled:

$$Value_{normalized} = (accumulator - A_{min}) / (A_{max} - A_{min})$$
$$Value_{DSP} = minimumValue + (Value_{normalized} \times (maximumValue - minimumValue))$$